UAS PSS

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```
library(MASS)
```

```
## Warning: package 'MASS' was built under R version 4.4.3
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.4.3
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.4.3
```

```
##
## Attaching package: 'dplyr'
```

```
## The following object is masked from 'package:MASS':
##
##     select
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.4.3
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.4.3
```

```
##
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
##
##     recode
```

```
library(corrplot)
```

```
## Warning: package 'corrplot' was built under R version 4.4.3
```

```
## corrplot 0.95 loaded
```

```
# Membangkitkan data simulasi

set.seed(123)
n <- 200
X1 <- rnorm(n, 3, 0.5)
X2 <- rnorm(n, 20, 5)
X3 <- rnorm(n, 80, 10)

# Menentukan model dan parameter sebenarnya

# Model logistik
logit_p <- -5 + 2*X1 + 0.08*X2 + 0.03*X3
p <- 1 / (1 + exp(-logit_p))

# Parameter sebenarnya
beta_true <- c(
  beta0 = -5,
  beta1 = 2,
  beta2 = 0.08,
  beta3 = 0.03
)

# Hasil distribusi binomial
Y <- rbinom(n, 1, p)
Y
```

```
##   [1] 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
##  [38] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
##  [75] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [112] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [149] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## [186] 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1
```

```
# Membuat skenario A yaitu data dengan outlier (5 data ekstrem)
idx <- sample(1:n, 5)
X2[idx] <- X2[idx] + 100
X3[idx] <- X3[idx] - 80

data_A <- data.frame(Y, X1, X2, X3)
data_A
```

```
##     Y       X1         X2         X3
## 1   1 2.719762  30.994052  79.264440
## 2   1 2.884911  26.562065  68.313486
## 3   1 3.779354  18.674275  73.652517
## 4   1 3.035254  22.715970  79.711584
## 5   1 3.064644  17.928300  86.706960
## 6   1 3.857532  17.618766  63.494535
## 7   1 3.230458  16.056986  76.502458
## 8   1 2.367469  17.026914  87.564064
## 9   1 2.656574  28.254537  74.611908
## 10  1 2.777169  19.729859  82.272919
## 11  1 3.612041  20.596226  84.922286
## 12  0 3.179907  21.218437  82.678350
## 13  1 3.200386 126.162379   6.532577
## 14  1 3.055341  17.419681  78.772913
## 15  1 2.722079  15.037464  75.863235
## 16  1 3.893457  28.378485  53.568510
## 17  1 3.248925  17.794184  79.070590
## 18  1 2.016691  16.384670  84.302847
## 19  1 3.350678  13.818634  85.353988
## 20  1 2.763604  13.576421  74.447216
## 21  1 2.466088  17.130133  97.795029
## 22  1 2.891013  23.089929  82.864244
## 23  1 2.486998  25.549241  81.263159
## 24  1 2.635554  23.537942  92.722668
## 25  1 2.687480 118.181714  -7.184662
## 26  1 2.156653  20.298750  75.496614
## 27  1 3.418894  16.477018 103.974525
## 28  1 3.076687  16.413909  80.111292
## 29  1 2.430932  24.423252  96.335684
## 30  1 3.626907  14.922037  65.614934
## 31  1 3.213232  29.776470  78.094832
## 32  1 2.852464  19.548402  83.784239
## 33  1 3.447563  21.072694  83.000385
## 34  1 3.439067  16.307361  69.943637
## 35  1 3.410791  17.128057  80.192593
## 36  1 3.344320  13.414919  69.225793
## 37  1 3.276959  19.085373  87.127033
## 38  1 2.969044  22.094912  90.847751
## 39  1 2.847019  21.621522  57.750123
## 40  1 2.809764  16.092318  92.356935
## 41  1 2.652647  16.056890  67.589555
## 42  1 2.896041  17.489006  84.547693
## 43  1 2.367302  27.480303  86.599026
## 44  1 4.084478  14.313482  78.001102
## 45  1 3.603981  19.104742  73.548860
## 46  1 2.438446  29.511809  81.653210
## 47  1 2.798558  19.495126  84.388187
## 48  1 2.766672  13.200796  88.833028
## 49  1 3.389983  16.676153  59.476630
## 50  1 2.958315  22.427300  63.636207
## 51  1 3.126659  18.121986  94.304023
```

```
## 52  1 2.985727  17.190618  90.466288
## 53  1 2.978565  18.280414  84.352889
## 54  1 3.684301  20.452483  87.151784
## 55  1 2.887115  27.992544  89.171749
## 56  1 3.758235  19.557174  53.390772
## 57  1 2.225624  25.403997  91.102771
## 58  1 3.292307  23.153771  75.150124
## 59  1 3.061927  19.431801  82.306168
## 60  1 3.107971  12.335490  77.048422
## 61  1 3.189820  17.394413  88.719650
## 62  1 2.748838  17.550648  76.515276
## 63  1 2.833396  20.235772  85.185038
## 64  1 2.490712  26.500993  76.093150
## 65  1 2.464104  31.465395  69.072128
## 66  1 3.151764  27.737905  92.100105
## 67  1 3.224105  19.334245  87.409000
## 68  1 3.026502 111.217363  17.242622
## 69  1 3.461134  18.056101  80.651539
## 70  1 4.025042  20.446036  91.250027
## 71  1 2.754484  24.225065  99.754191
## 72  1 1.845416  24.812640  77.185179
## 73  1 3.502869  23.421547  66.770489
## 74  1 2.645400  13.023628  77.606484
## 75  1 2.655996  24.248215  77.859588
## 76  1 3.512786  17.767214  81.516805
## 77  1 2.857613  20.874014  97.123050
## 78  1 2.389641  20.372756  76.738561
## 79  1 3.090652  22.140834  83.730047
## 80  1 2.930554  20.123375  77.723159
## 81  1 3.002882  11.662625  80.204507
## 82  1 3.192640  23.682480  83.140577
## 83  1 2.814670  21.930133  93.282147
## 84  1 3.322188  18.671742  81.213184
## 85  1 2.889757  20.590723  87.128423
## 86  1 3.165891  20.670193  87.788600
## 87  1 3.548420  21.105097  89.147733
## 88  1 3.217591  28.204231  74.256054
## 89  1 2.837034  18.904748  96.268812
## 90  1 3.574404  20.840327  76.190433
## 91  1 3.496752  25.841919  78.942158
## 92  1 3.274198  25.270905  94.040503
## 93  1 3.119366  25.726316  92.940839
## 94  1 2.686047  17.112660  69.100081
## 95  1 3.680326  30.012414  71.269290
## 96  1 2.699870  20.333504  66.419209
## 97  1 4.093666  29.334259  81.818472
## 98  1 3.766305  13.245487  81.648409
## 99  1 2.882150  20.104918  83.641147
## 100 1 2.486790  26.249573  85.521577
## 101 1 2.644797  16.423789  73.981072
## 102 1 3.128442  16.236555  70.063014
## 103 1 2.876654  15.307306  90.267851
```

```
## 104 1 2.826229  14.737434  87.510613
## 105 1 2.524191  17.814202  64.908335
## 106 1 2.977486  21.655896  79.048525
## 107 1 2.607548   9.928948  71.040522
## 108 1 2.166029  21.059902  59.292489
## 109 1 2.809887  26.183375  81.501201
## 110 1 3.459498  30.187870  79.207883
## 111 1 2.712327  26.505880  79.026307
## 112 1 3.303982  23.783874  82.161525
## 113 1 2.191059  11.366348  88.824652
## 114 1 2.972219  16.992466  82.055975
## 115 1 3.259704  18.239768  73.835642
## 116 1 3.150577  23.517620  72.652007
## 117 1 3.052838  19.471643  78.681972
## 118 1 2.679647  13.706757  83.100170
## 119 1 2.575148  28.422179  69.603196
## 120 1 2.487936  24.556956  78.156911
## 121 1 3.058823  21.187151  89.672673
## 122 1 2.526263  26.090543  78.917199
## 123 1 2.754721  13.306129  73.015793
## 124 1 2.871954  23.304101  77.240548
## 125 1 3.921931  17.385438  91.146485
## 126 1 2.674025  23.418728  85.500440
## 127 1 3.117693 119.695890  12.366758
## 128 1 3.038980  23.164804  81.390979
## 129 1 2.519072  26.677588  84.102751
## 130 1 2.964346  20.036450  74.415431
## 131 1 3.722275  25.087793  86.053707
## 132 1 3.225752  14.057830  74.936665
## 133 1 3.020616  16.391978  65.794345
## 134 1 2.788752  27.596089  81.279930
## 135 1 1.973376  21.886940  99.458512
## 136 1 3.565669   9.738886  88.009143
## 137 1 2.269680  13.179813  91.652534
## 138 1 3.369974  18.996095  83.588557
## 139 1 3.954552  24.328897  73.914428
## 140 1 2.278053  19.490584  77.977591
## 141 1 3.350892  23.120937  77.267519
## 142 1 2.868901  24.795027  75.313002
## 143 1 2.213928  28.355274  87.041673
## 144 1 2.242666  20.280084  68.026365
## 145 1 2.199232  19.740090  88.663661
## 146 1 2.734547  11.233813  88.641525
## 147 1 2.269122  20.496638  68.013776
## 148 1 3.343958  17.140750  86.394920
## 149 1 4.050054  15.129952 104.302267
## 150 1 2.356485  19.100469  74.427845
## 151 1 3.393869  25.074716  88.449042
## 152 1 3.384521  10.036258  72.177982
## 153 1 3.166101  17.863604  91.107114
## 154 1 2.495812  20.583186  82.498247
## 155 1 2.940274  15.533962  96.519154
```

```
## 156 1 2.859802  21.669515  65.410293
## 157 1 3.281495  22.057150  79.487021
## 158 1 2.813781  19.834819  74.730748
## 159 1 3.488487   7.670509  78.027351
## 160 1 2.812710  32.857291  73.704213
## 161 1 3.526356  18.973504  71.661564
## 162 1 2.475411  23.255966  85.787224
## 163 1 2.369922  21.368832  69.124193
## 164 1 4.620520  25.123366  94.840309
## 165 1 2.791571  24.088297  68.137934
## 166 1 3.149114  18.951034  81.010792
## 167 1 3.318285  21.890839  85.329893
## 168 1 2.758110  15.272956  85.867353
## 169 1 3.258431  24.284615  76.982533
## 170 1 3.184482  17.694808  80.795020
## 171 1 2.892310  32.083867  89.612642
## 172 1 3.032647  11.744756  65.435341
## 173 1 2.982966  17.680064  72.182603
## 174 1 4.064226  24.126899  83.204023
## 175 1 2.629332  22.550663  75.552180
## 176 1 2.452002  17.052595  93.700040
## 177 1 3.018894  15.016096  86.732539
## 178 1 3.155240  20.722379  80.721668
## 179 1 3.218262  19.928463  64.922427
## 180 1 2.770817  11.048594  80.261002
## 181 1 2.468337  20.172755  76.835841
## 182 1 3.631593 120.951152  -1.023465
## 183 1 2.825175  20.873632  68.184408
## 184 1 2.567244  14.724915  84.986580
## 185 1 2.881860  22.380666  69.610436
## 186 1 2.901412  26.892851  77.737780
## 187 1 3.554960  22.281182  83.814258
## 188 1 3.042369  14.322058  72.164842
## 189 1 3.377027  17.821773  85.829914
## 190 1 2.750354  21.730518  66.834896
## 191 0 3.107223  16.764772  51.902253
## 192 1 2.837657   9.211768  84.649680
## 193 1 3.047292  24.421254  88.405398
## 194 1 2.552318  15.852612  77.141546
## 195 1 2.344599  17.132199  85.041263
## 196 1 3.998607  27.519503  68.440835
## 197 1 3.300354  16.129275  78.728514
## 198 1 2.374364  24.228658  60.584816
## 199 1 2.694417  13.696586  91.811809
## 200 1 2.407260  18.227288  98.599109
```

```
# Estimasi model
model_A <- glm(Y ~ X1 + X2 + X3, data = data_A, family = binomial)
summary(model_A)
```

```
##
## Call:
## glm(formula = Y ~ X1 + X2 + X3, family = binomial, data = data_A)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -5.38757    8.48405  -0.635   0.5254
## X1          -0.37585    1.58781  -0.237   0.8129
## X2           0.10271    0.08721   1.178   0.2389
## X3           0.12320    0.07251   1.699   0.0893 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 22.401  on 199  degrees of freedom
## Residual deviance: 18.790  on 196  degrees of freedom
## AIC: 26.79
##
## Number of Fisher Scoring iterations: 8
```

```
# Evaluasi model
se_A <- summary(model_A)$coefficients[,2]
se_A
```

```
## (Intercept)          X1          X2          X3
##  8.48405159  1.58780635  0.08720737  0.07250749
```

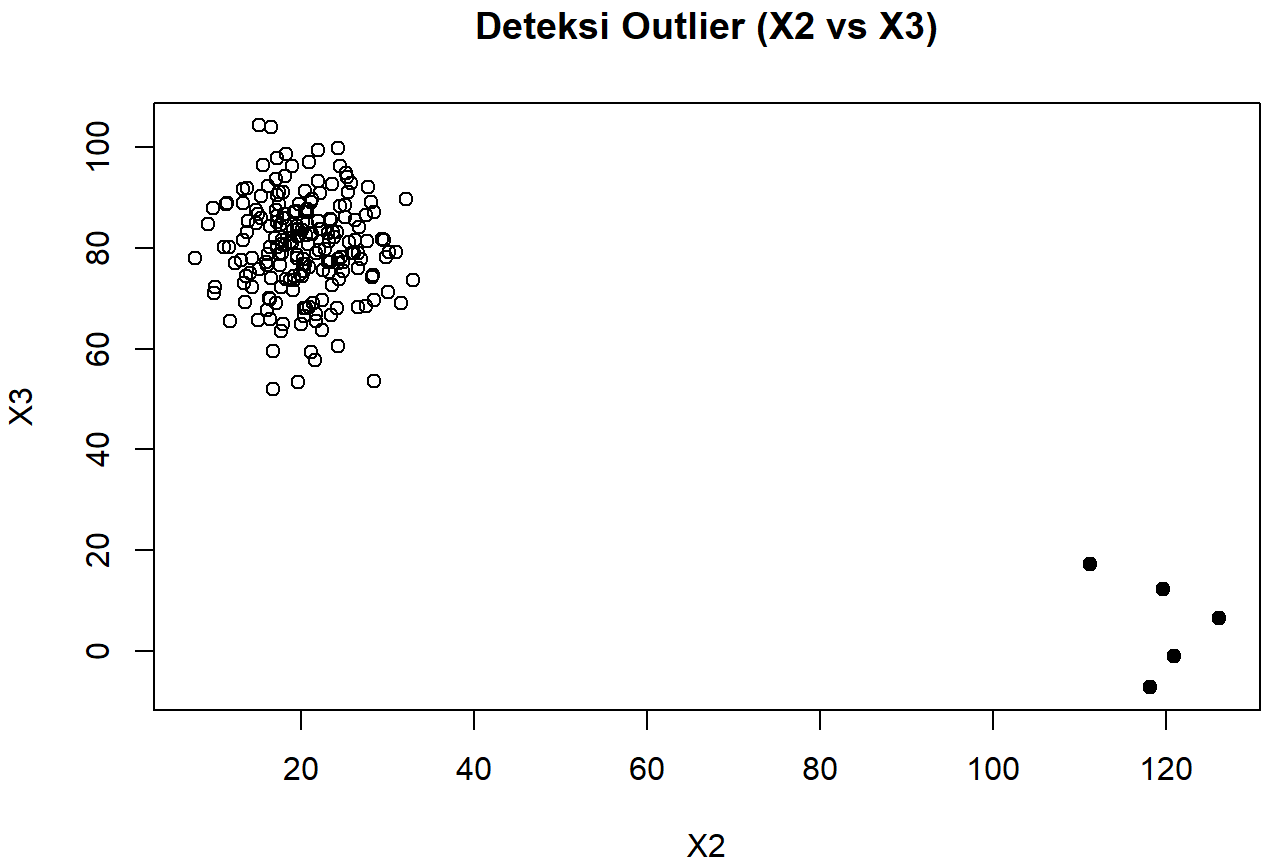
```
# Akurasi klasifikasi
pred_prob <- predict(model_A, type = "response")
pred_class <- ifelse(pred_prob > 0.5, 1, 0)

accuracy <- mean(pred_class == data_A$Y)
accuracy
```

```
## [1] 0.99
```

```
# Mendeteksi outlier berpengaruh
cook <- cooks.distance(model_A)

# Observasi dengan pengaruh besar
which(cook > (4/n))
```

```
##  12  56 191
##  12  56 191
```

```
plot(data_A$X2, data_A$X3,
     main="Deteksi Outlier (X2 vs X3)",
     xlab="X2", ylab="X3")

points(data_A$X2[idx], data_A$X3[idx], pch=19)
```



```
# Membuat skenario B yaitu adanya multikolinearitas pada data dan membangkitkan data simulasi
set.seed(124)
n <- 200

# Variabel utama
X1 <- rnorm(n, 3, 0.5)

# Multikolinearitas kuat
X2 <- 5*X1 + rnorm(n, 0, 0.1)
X3 <- 10*X1 + rnorm(n, 0, 0.1)

# Model logistik
logit_p <- -5 + 2*X1 + 0.08*X2 + 0.03*X3
p <- 1 / (1 + exp(-logit_p))

Y <- rbinom(n, 1, p)

data_B <- data.frame(Y, X1, X2, X3)
data_B
```

```
##     Y       X1        X2       X3
## 1   1 2.307465 11.700676 23.06311
## 2   1 3.019162 15.058143 30.20454
## 3   1 2.618485 13.075801 26.21157
## 4   1 3.106153 15.527550 30.92223
## 5   1 3.712769 18.615160 37.06865
## 6   1 3.372240 16.988022 33.93470
## 7   1 3.350115 16.644545 33.50847
## 8   1 2.885323 14.488580 28.81451
## 9   1 3.098547 15.410772 30.85251
## 10  1 3.603577 17.969110 36.07026
## 11  1 3.159168 15.786733 31.49068
## 12  1 2.288101 11.471296 22.87992
## 13  0 2.797455 13.949784 27.88803
## 14  1 3.497693 17.409681 34.97259
## 15  1 3.479409 17.295411 34.76319
## 16  1 3.459044 17.032718 34.73741
## 17  1 2.924515 14.599593 29.29174
## 18  0 2.388466 11.915736 23.92761
## 19  1 2.565588 12.864321 25.79630
## 20  1 2.478757 12.405355 24.73293
## 21  0 2.448181 12.316636 24.33854
## 22  1 3.222093 16.261994 32.25811
## 23  1 2.897525 14.221645 29.06234
## 24  1 3.837816 19.120263 38.48180
## 25  1 2.934339 14.629037 29.33257
## 26  1 2.900059 14.457672 28.80364
## 27  1 3.027456 15.152111 30.13508
## 28  1 2.658917 13.547367 26.49222
## 29  1 2.636148 13.166400 26.33711
## 30  0 2.569048 12.948546 25.80253
## 31  0 2.981238 14.870406 29.78833
## 32  1 2.184288 10.997214 21.78999
## 33  1 3.088583 15.672791 30.69319
## 34  0 2.993750 14.961191 29.88919
## 35  1 2.802841 14.300332 28.16703
## 36  1 3.175781 15.738028 31.72065
## 37  1 3.439384 17.204563 34.34064
## 38  1 3.102327 15.334314 31.18300
## 39  1 2.556310 12.760495 25.39828
## 40  1 2.761392 13.776581 27.66771
## 41  1 2.866130 14.269465 28.69606
## 42  1 3.792930 18.950606 37.79976
## 43  1 3.023450 15.129387 30.09706
## 44  1 3.178248 15.964613 31.79217
## 45  0 2.939310 14.919160 29.37022
## 46  1 2.981954 14.967450 29.76485
## 47  1 2.509426 12.563582 25.05060
## 48  1 2.782870 13.993840 27.81902
## 49  0 2.966258 14.935049 29.60961
## 50  1 3.490947 17.309469 34.85756
## 51  1 2.768974 13.894285 27.83083
```

```
## 52  1 2.887450 14.354170 28.85642
## 53  0 2.576776 12.964679 25.70567
## 54  1 3.036523 15.203557 30.21738
## 55  1 2.862482 14.284652 28.54882
## 56  1 2.806787 13.887109 28.05293
## 57  1 2.976898 14.934224 29.80601
## 58  1 2.587053 12.949909 25.78799
## 59  0 2.572983 12.929074 25.58405
## 60  1 3.059368 15.303740 30.41869
## 61  1 3.141798 15.710845 31.48521
## 62  1 3.965043 19.798174 39.73815
## 63  1 2.429736 12.125445 24.26300
## 64  0 2.338941 11.599915 23.60800
## 65  1 3.614416 18.099453 36.25085
## 66  0 2.725772 13.621890 27.22665
## 67  1 2.936996 14.640515 29.32282
## 68  1 3.343859 16.723174 33.41825
## 69  1 3.352600 16.731545 33.55657
## 70  1 3.400739 17.190981 34.08360
## 71  1 2.768558 13.703859 27.89599
## 72  1 2.557723 12.858333 25.45818
## 73  1 2.184536 10.806488 21.96119
## 74  1 3.281115 16.312802 32.77571
## 75  0 2.837054 14.214655 28.51839
## 76  1 3.120688 15.623986 31.10783
## 77  1 3.542935 17.754315 35.35179
## 78  1 3.953531 19.750159 39.66874
## 79  1 3.406464 16.945714 34.02899
## 80  1 3.247444 16.162087 32.32785
## 81  1 2.952426 14.596150 29.52198
## 82  1 3.569390 17.600640 35.75432
## 83  1 3.251157 16.355523 32.42734
## 84  1 2.742293 13.831589 27.40311
## 85  0 1.765805  8.620126 17.61769
## 86  1 2.563724 12.918895 25.75110
## 87  1 3.482044 17.390117 34.80339
## 88  1 3.455398 17.317356 34.43878
## 89  1 3.962904 19.843359 39.66008
## 90  1 2.848547 14.242896 28.34426
## 91  1 2.472646 12.296001 24.75856
## 92  0 3.209058 16.133375 32.01664
## 93  1 3.350636 16.692017 33.53640
## 94  1 3.123379 15.514009 31.14169
## 95  1 3.232148 16.357367 32.29054
## 96  1 2.802266 13.884167 27.92196
## 97  1 3.356535 16.791981 33.44280
## 98  1 3.592506 17.993238 35.84055
## 99  1 2.044425 10.223080 20.46926
## 100 1 3.557465 17.717789 35.55848
## 101 1 2.765350 13.821851 27.60140
## 102 1 2.425999 12.219045 24.25459
## 103 1 3.551999 18.041038 35.70602
```

```
## 104 1 2.855375 14.309634 28.57033
## 105 1 2.861336 14.202268 28.69363
## 106 1 2.562109 12.731547 25.55151
## 107 1 3.001640 15.028014 29.92895
## 108 1 3.132796 15.648498 31.31101
## 109 1 2.937988 14.598344 29.42888
## 110 1 2.935537 14.579941 29.21208
## 111 1 3.149556 15.903457 31.49012
## 112 1 3.009150 15.039674 30.03661
## 113 1 2.424969 12.113675 24.28182
## 114 1 2.785302 13.811642 27.78595
## 115 1 3.425001 17.275628 34.23273
## 116 1 2.892585 14.565534 28.93483
## 117 1 2.691293 13.355579 26.85640
## 118 1 3.054714 15.324370 30.61714
## 119 1 2.646744 13.215611 26.45858
## 120 1 3.771488 18.736758 37.64063
## 121 1 3.175401 15.938313 31.88049
## 122 1 3.586310 18.096250 36.04141
## 123 1 2.983313 14.958179 29.71343
## 124 1 3.180722 15.915621 31.87539
## 125 1 3.522788 17.579252 35.38005
## 126 1 3.051606 15.092254 30.42154
## 127 1 3.165898 15.937203 31.74267
## 128 1 2.215721 11.004382 22.10914
## 129 1 3.220135 16.117825 32.36536
## 130 1 3.940936 19.767200 39.27534
## 131 1 3.674553 18.258855 36.67871
## 132 1 2.182459 10.860480 21.83237
## 133 1 2.642234 13.415052 26.36272
## 134 1 2.541753 12.622335 25.36260
## 135 0 1.819487  9.016320 18.28479
## 136 1 3.764225 18.894110 37.56952
## 137 1 2.685106 13.287285 26.78532
## 138 1 2.465954 12.124991 24.59387
## 139 1 3.622430 18.136294 36.25706
## 140 1 3.229938 16.084783 32.26069
## 141 1 3.083853 15.252721 30.91328
## 142 1 2.862498 14.320554 28.53965
## 143 1 3.246161 16.124231 32.40099
## 144 1 3.669877 18.397229 36.82519
## 145 1 2.761628 13.945037 27.67048
## 146 1 2.251602 11.070768 22.53081
## 147 1 2.399288 12.046918 23.86619
## 148 1 2.796269 14.058945 28.00825
## 149 1 2.526126 12.714810 25.16126
## 150 1 3.425833 17.182651 34.21736
## 151 1 2.705784 13.461279 27.07017
## 152 1 2.318763 11.448871 23.12489
## 153 1 2.775181 13.935461 27.70793
## 154 0 2.678337 13.469559 26.61002
## 155 1 2.744627 13.705853 27.36679
```

```
## 156 1 3.128601 15.708714 31.19705
## 157 1 2.790222 13.994637 27.92067
## 158 0 2.523886 12.555900 25.25621
## 159 1 3.256197 16.479404 32.61290
## 160 1 2.127866 10.705415 21.25631
## 161 1 2.751244 13.750303 27.56311
## 162 1 3.382793 16.928450 33.75440
## 163 1 3.250697 16.259547 32.44867
## 164 1 3.165956 15.708068 31.66060
## 165 1 2.954462 14.865441 29.58159
## 166 1 2.671659 13.173167 26.70099
## 167 1 2.419576 12.132895 24.16176
## 168 1 3.827966 19.148391 38.36996
## 169 0 2.625395 13.163738 26.21866
## 170 1 3.310960 16.642818 33.00360
## 171 1 2.264097 11.243480 22.63052
## 172 1 3.399098 16.675957 33.92825
## 173 1 3.139731 15.604857 31.42480
## 174 1 3.328937 16.734956 33.17256
## 175 1 3.296863 16.542846 32.82608
## 176 0 2.711037 13.561520 27.26518
## 177 1 3.238031 16.367151 32.51147
## 178 0 2.406183 11.987381 23.98494
## 179 1 3.358844 16.847194 33.72186
## 180 1 3.067349 15.346914 30.74079
## 181 1 3.711040 18.401946 37.21673
## 182 0 3.020365 15.091073 30.31772
## 183 1 2.830795 14.230209 28.32287
## 184 0 2.720942 13.581332 27.22050
## 185 1 3.513907 17.640601 35.10097
## 186 1 2.887265 14.435221 28.87403
## 187 1 2.384185 11.938475 23.92452
## 188 1 2.620878 13.001075 26.17132
## 189 1 3.542759 17.919042 35.53730
## 190 1 2.475451 12.249545 24.82464
## 191 0 3.145200 15.695921 31.42780
## 192 0 2.776213 13.962768 27.81180
## 193 1 2.890570 14.405425 28.81622
## 194 1 2.197278 10.985092 22.00429
## 195 1 3.037420 15.169513 30.27212
## 196 1 2.739041 13.673046 27.33488
## 197 0 2.967549 14.778838 29.73671
## 198 1 3.525685 17.632052 35.37084
## 199 1 2.128921 10.567550 21.47958
## 200 1 3.866274 19.133480 38.44612
```

```
# Estimasi model
model_B <- glm(Y ~ X1 + X2 + X3, data = data_B, family = binomial)
summary(model_B)
```

```
##
## Call:
## glm(formula = Y ~ X1 + X2 + X3, family = binomial, data = data_B)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -3.934      1.553  -2.534   0.0113 *
## X1            27.986     27.604   1.014   0.3107
## X2            -2.764      2.245  -1.231   0.2183
## X3            -1.212      2.505  -0.484   0.6286
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 154.55  on 199  degrees of freedom
## Residual deviance: 138.19  on 196  degrees of freedom
## AIC: 146.19
##
## Number of Fisher Scoring iterations: 5
```

```
# Evaluasi model
se_B <- summary(model_B)$coefficients[,2]
se_B
```

```
## (Intercept)          X1          X2          X3
##    1.552574   27.603948    2.244991    2.504775
```

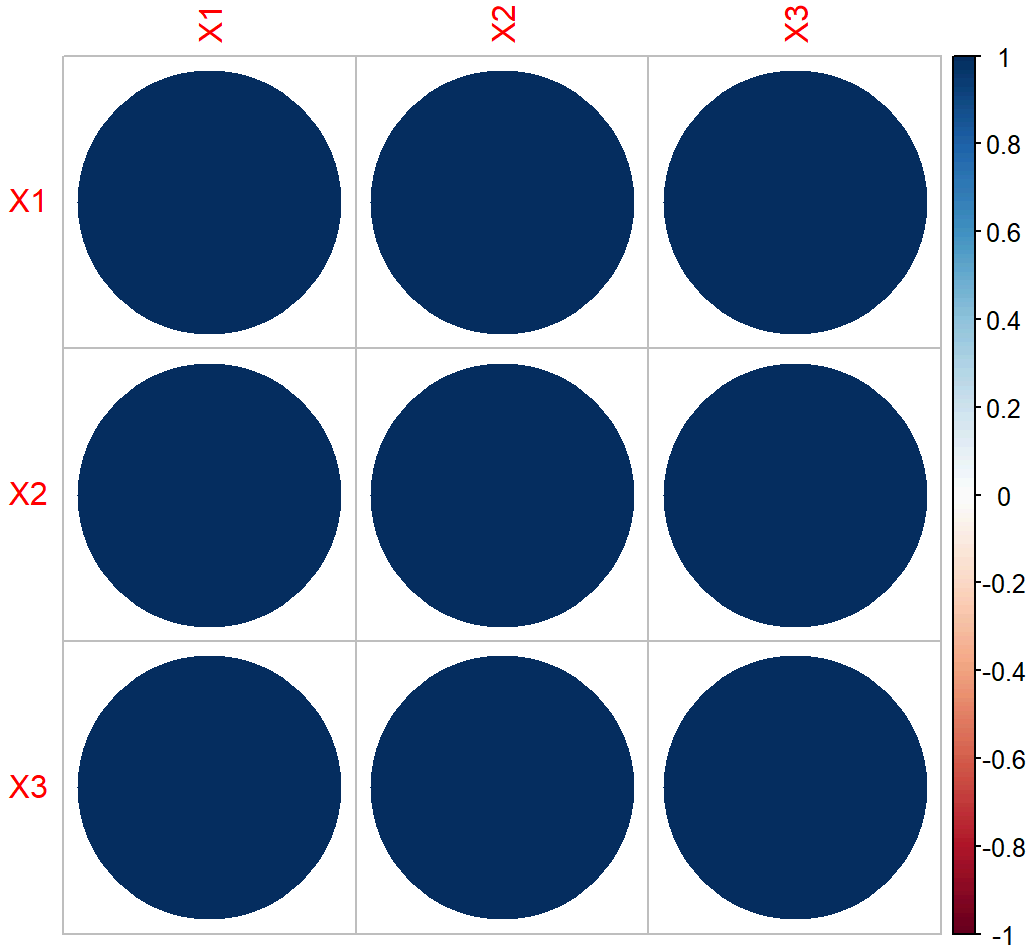
```
# Akurasi klasifikasi
pred_prob <- predict(model_B, type = "response")
pred_class <- ifelse(pred_prob > 0.5, 1, 0)

accuracy <- mean(pred_class == data_B$Y)
accuracy
```

```
## [1] 0.875
```

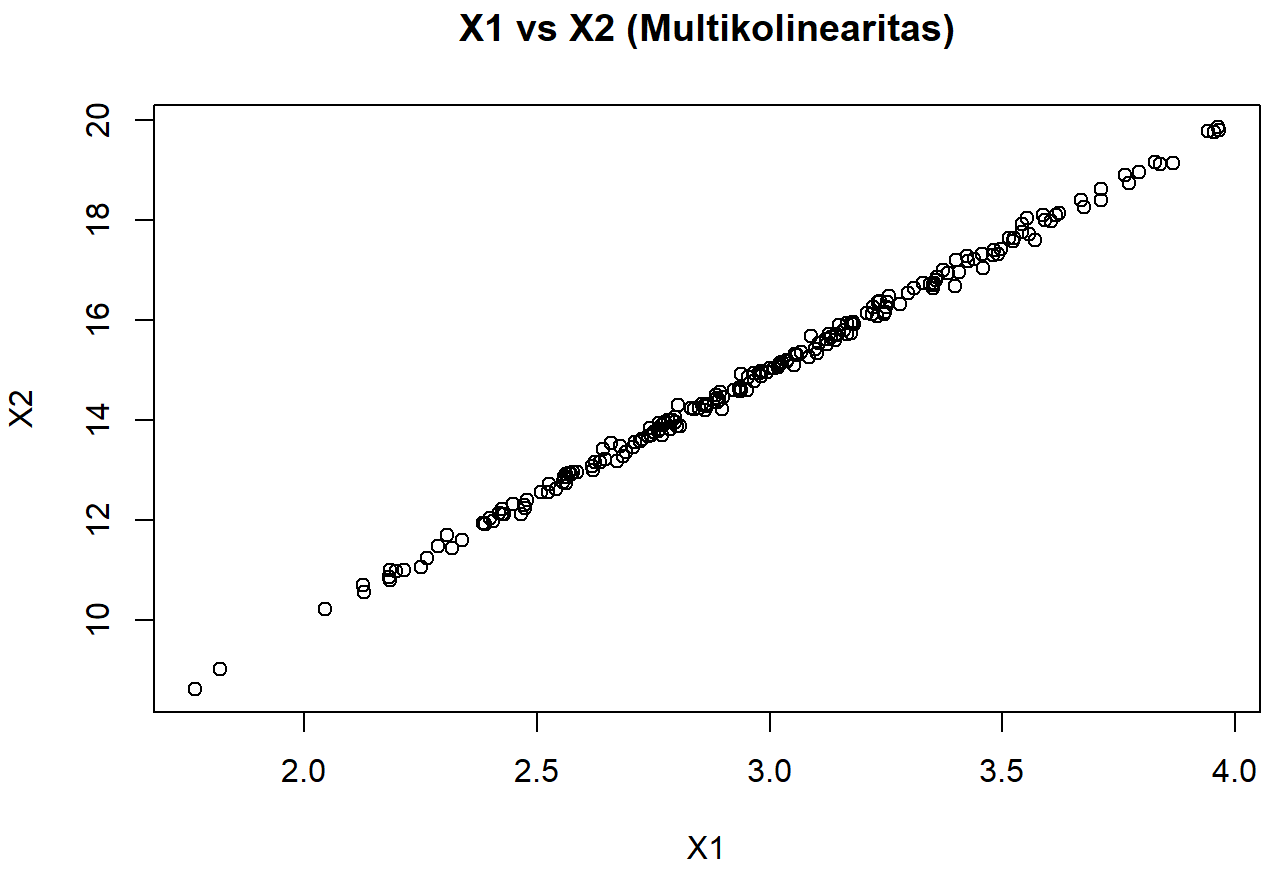
```
# Visualisasi
cor_matrix <- cor(data_B[, c("X1","X2","X3")])
cor_matrix
```

```
##           X1        X2        X3
## X1 1.0000000 0.9989545 0.9997970
## X2 0.9989545 1.0000000 0.9987784
## X3 0.9997970 0.9987784 1.0000000
```

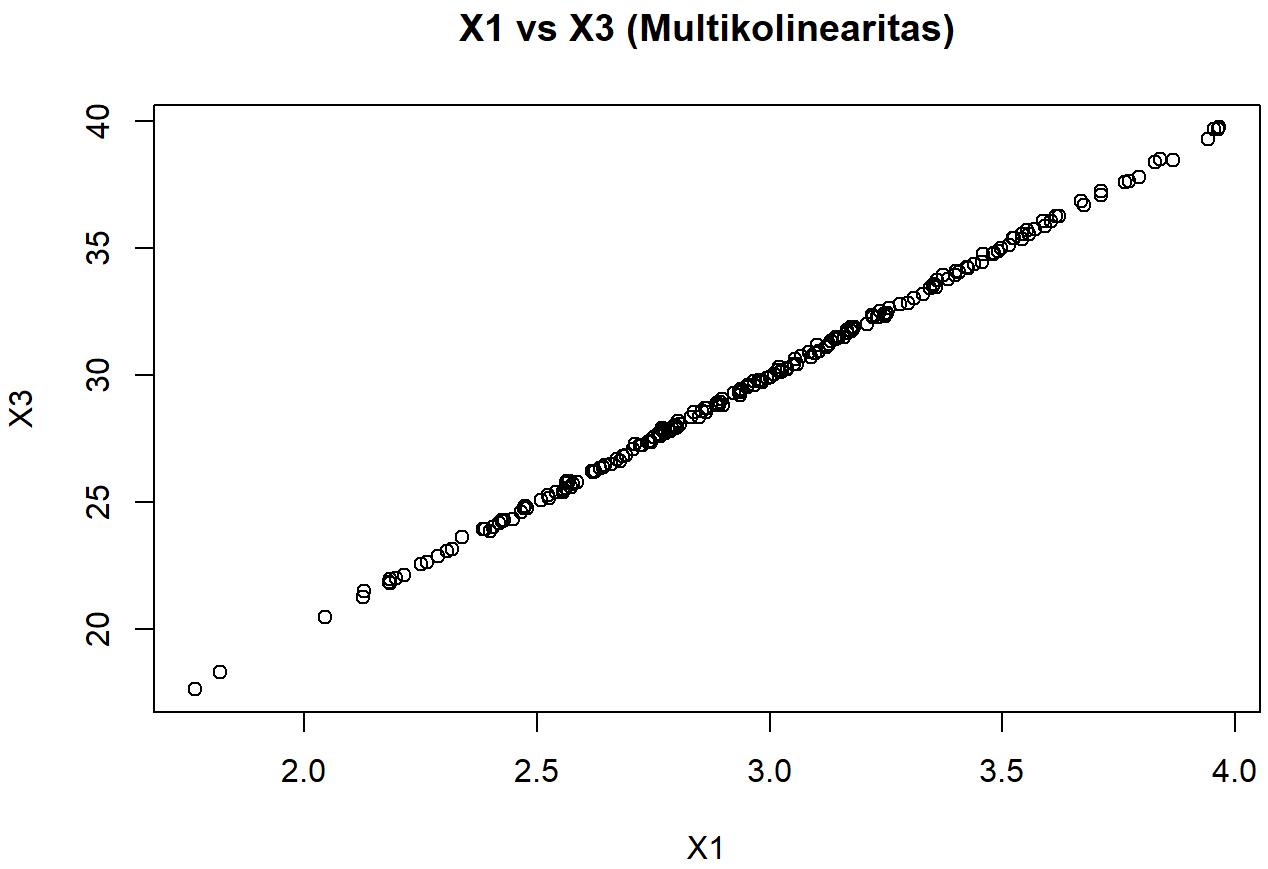
```
corrplot(cor_matrix, method="circle")
```




```
plot(data_B$X1, data_B$X2,
     main="X1 vs X2 (Multikolinearitas)",
     xlab="X1", ylab="X2")
```




```
plot(data_B$X1, data_B$X3,
     main="X1 vs X3 (Multikolinearitas)",
     xlab="X1", ylab="X3")
```


X1 vs X3 (Multikolinearitas)
X3
20
25
30
35
40
2.0
2.5
3.0
3.5
4.0
X1